Test – Trigonometric Functions November 18, 2024

20.5 / 44 What happened?

1. (a) Convert 255° into radian measure. (b) Convert $\frac{2\pi}{9}$ into degree measure.

2 1 $225° \times \frac{\pi}{180°} = \frac{5}{4}\pi$ ✓ $\frac{2\pi}{9} \times \frac{180°}{\pi} = 40°$ ✓

1

oops

2. For each of the following calculate the exact value. Show your work!

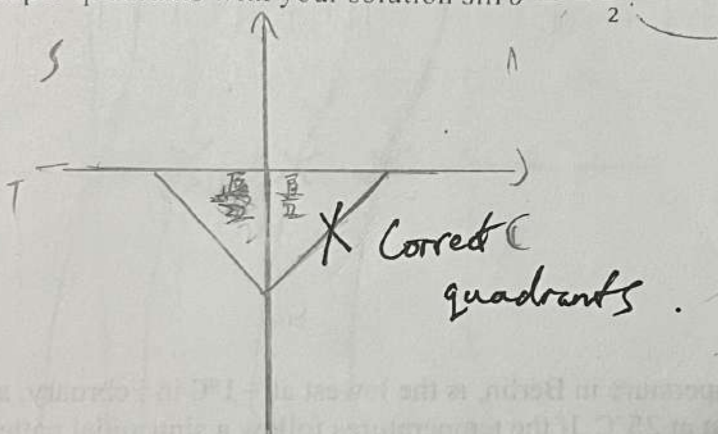
(a) $\tan\left(\frac{7\pi}{6}\right)$ (b) $\csc\left(\frac{3\pi}{4}\right)$

$\frac{7}{6}\pi \times \frac{180}{\pi} = 210°$ $\csc = \frac{1}{\sin}$

2 2 $\frac{3}{4}\pi \times \frac{180°}{\pi} = 135°$

2 $\tan \cancel{30°} = \frac{\sqrt{3}}{3}$ ✓ $\frac{1}{\sin 135°} = \sqrt{2}$ ✓

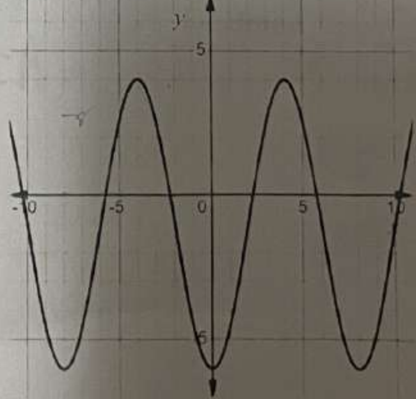
3. Determine the exact value(s) for θ where $0 \leq \theta \leq 2\pi$. Draw a sketch showing them in the proper quadrants with your solution $\sin\theta = -\frac{\sqrt{3}}{2}$.



1 5 ✗ Correct quadrants. $\theta = ?$

PF

4. Determine an equation for the following sinusoidal curve. Show all calculations along with your final equation.



max 4

min -6

2 4 period $2\pi \div 12 = \frac{1}{6}\pi$

$A = \frac{4-(-6)}{2} = 5$ ✓

$C = \frac{4-6}{2} = -1$ ✓

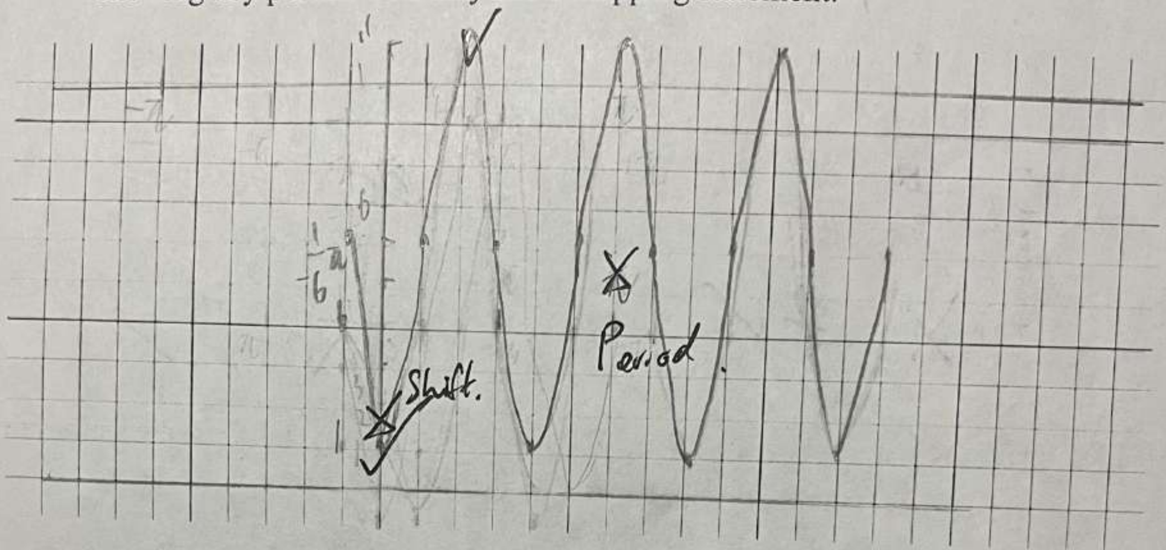
$y = 5\sin(\cancel{\tfrac{1}{6}x}) - 1$

5. (a) Complete the table for the function $y = -5\sin\left(2\theta + \frac{\pi}{3}\right) + 6$:

Factored Function	Amplitude	Period	Phase Shift	Axis of Curve	Range
$y = \sin x$? PF. On other page.	5 ✓	π ✓	$\frac{\pi}{6}$ left, 6 up	$y = \cancel{6}$ PF.	[illegible]

5 /6

(b) Graph the above function on the grid below, showing **two cycles**, clearly showing key points. You may use a mapping statement.



2 /4

6. The average temperature in Berlin, is the lowest at -1°C in February, and is the highest in August at 25°C. If the temperatures follow a sinusoidal pattern, then determine an equation for the average temperature, starting in January as month number 1.

$A = \frac{25-(-1)}{2} = 13$ ✓

$C \quad \frac{-1+25}{2} = 12$

$2\pi \div 12 = \frac{1}{6}\pi$

$y = 13\sin\left(\frac{1}{6}\pi x\right) + 12$

PF. Define x and y?

3 /4

7. The point $(-3, -7)$ is on the terminal arm of an angle θ in standard position. Determine the radian value of the angle to 2 decimal places, in the interval $0 \le \theta \le 2\pi$. Include a sketch in the correct quadrant.

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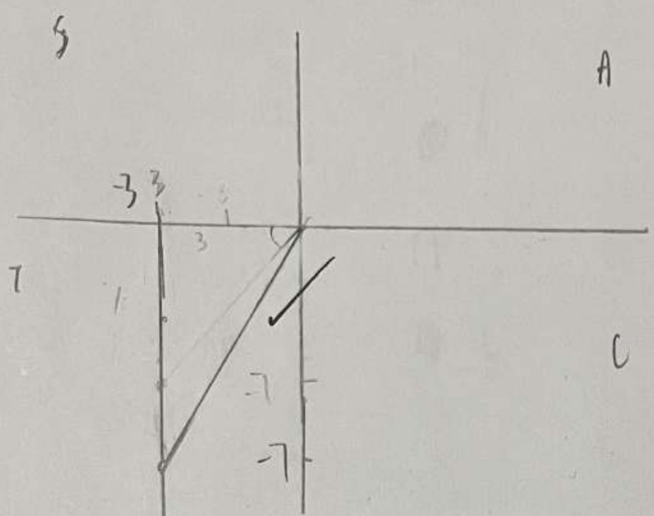

$\cos^{-1}\left(\frac{3}{7}\right) = \cancel{64.62}^{\circ}$

8. Using the graph of $y = \tan x$, **explain** how it can be used to graph of $y = \cot x$.

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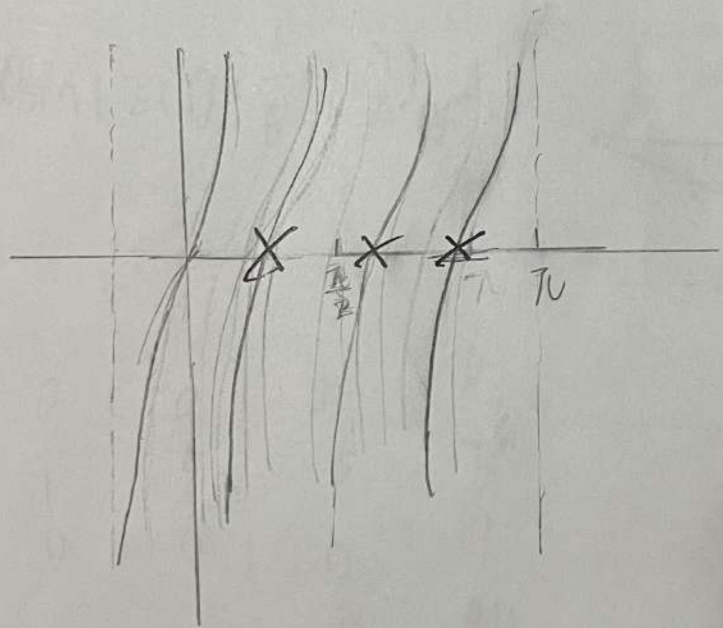
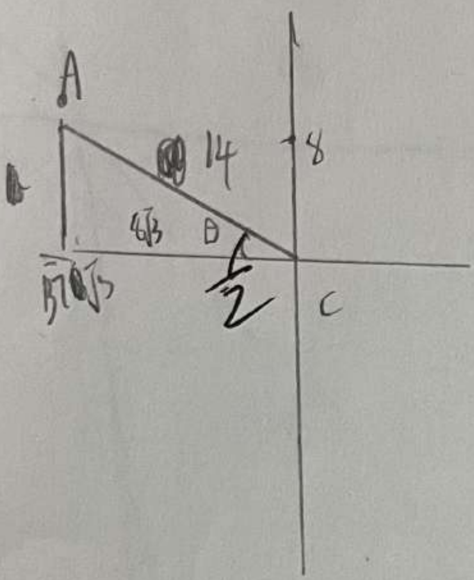


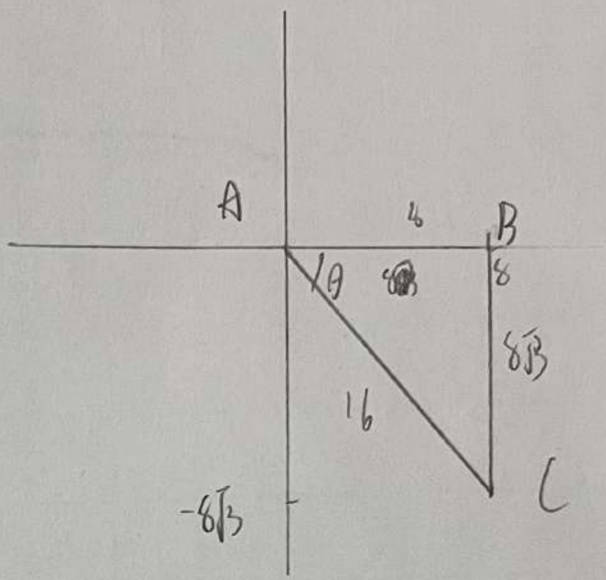
$\cot x = \frac{1}{\tan x}$

find the zeros of tan x and ~~reflect~~ the part of the graph

make sure that ~~tan x~~ do not touch the vertical line of (zeros, 0)

9. Find the sine of the angle formed by two lines that start at the origin of the Cartesian plane if one line passes through the point $(-7\sqrt{3},\ 7)$ and the other line passes through the point $(8, -8\sqrt{3})$



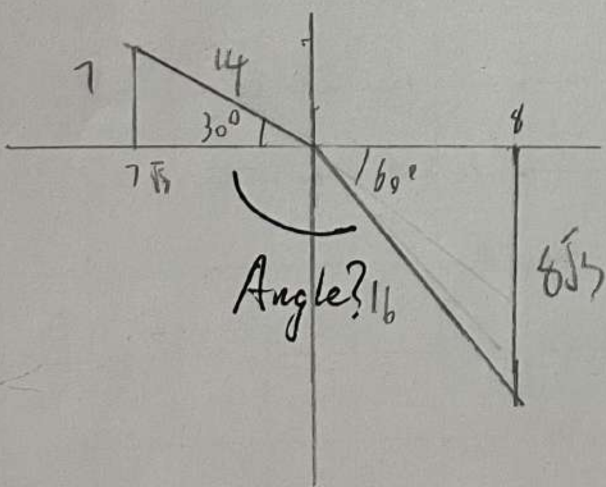


$\sqrt{196} = 14$

$AC = \sqrt{AB^2 + BC^2} = \sqrt{\cancel{256}}$ ~~16~~

$AC = \sqrt{AB^2 + BC^2} = \sqrt{256} = 16$

4 $\sin^{-1}\theta = \sin^{-1}\frac{7}{14} = 30^\circ$ Radians? $\sin^{-1}\theta = \sin^{-1}\frac{8\sqrt{3}}{16} = 60^\circ$ Radians?



4 ***Marks will be awarded for use of proper form throughout the assessment.***